ITERATIVE ROOT-FINDING METHODS AND THE POLYNOMIAL FACTORIZATION PROCEDURE

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Abstract. This paper implements and analyzes Newton's method, the secant method, and the bisection method to estimate real roots of univariate polynomials. We compare convergence rates, basins of attraction, and failure modes through theoretical and computational analysis. Newton's method achieves quadratic convergence ($q = 2$) for simple roots and linear convergence ($q = 1 - 1/k$) for roots of multiplicity k, whereas the bisection method guarantees linear convergence ($q = 1$) across any sign-changing bracket, and the secant method achieves superlinear convergence of order equal to the golden ratio ($\phi = \frac{1+\sqrt{5}}{2} \approx 1.618$). Finally, by combining Cauchy root bounds, critical-point partitioning, and inflection-guided Newton initialization, we introduce the **Polynomial Factorization Procedure (PFP)**—a recursive algorithm that deterministically isolates and approximates all real roots of any polynomial along with their exact multiplicities.

1. Introduction

A recurring problem in mathematics and scientific computing is to find the roots of polynomials. Computer software systems usually look for roots numerically rather than symbolically, through iterative estimation methods that converge closer and closer to the actual root with each step. We can compare root-finding methods by how fast they converge and by their failure cases; often, choosing the appropriate method involves a trade-off between fast convergence and robustness.

In this paper, we investigate three foundational iterative root-finding methods for polynomials in one variable: Newton's method, the secant method, and the bisection method. For consistency across our initial benchmarks, we use the polynomial

$$f(x) = \frac{1}{8}x^2 - \frac{1}{2},$$

whose positive root is $\alpha = 2$.

1.1. Newton's Method

Newton's method accepts an initial guess x_0, evaluates $f(x_0)$ and $f'(x_0)$, and intersects the tangent line at $(x_0, f(x_0))$ with the x-axis to form the next iterate $x_{n+1} = x_n - f(x_n)/f'(x_n)$. Figure 1 and Table 1 show Newton's method on $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ starting from $x_0 = 4$, alongside its failure mode when $f'(x_0) = 0$.

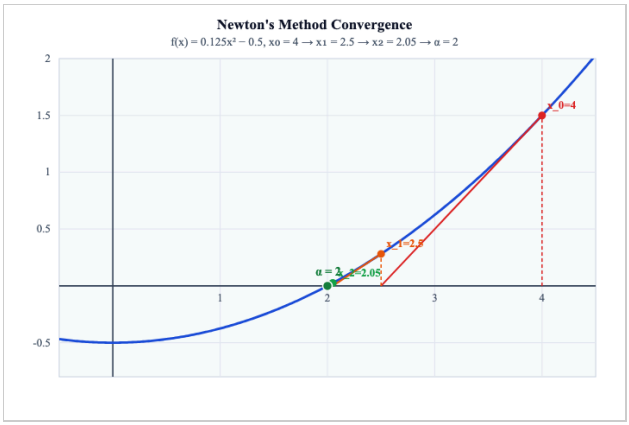


Figure 1a. Quadratic convergence of Newton's method from $x_0 = 4$ to $\alpha = 2$.

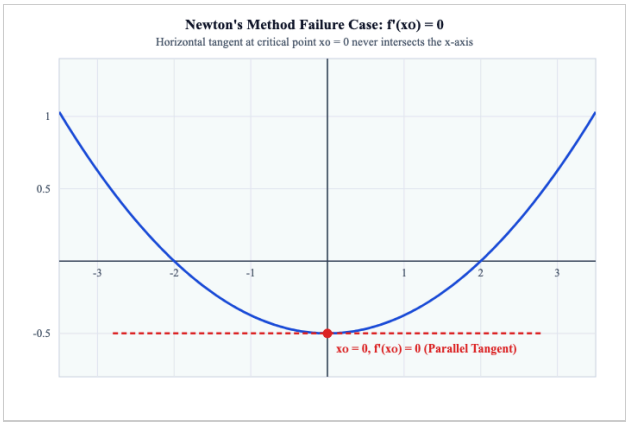


Figure 1b. Newton failure at critical point $x_0 = 0$ where $f'(x_0) = 0$.

Table 1. Newton's Method Iterations for $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ ($x_0 = 4$).

Step n	x_n	α	$e_n = x_n - \alpha$	$f(x_n)$
0	4.000000	2	2.000000	1.500000
1	2.500000	2	0.500000	0.281250
2	2.050000	2	0.050000	0.025312
3	2.000610	2	0.000610	0.000305
4	2.000000	2	0.000000	0.000000

1.2. Secant Method

The Secant method approximates the derivative using two initial guesses x_0 and x_1, computing the x-intercept of the secant line through $(x_{n-1}, f(x_{n-1}))$ and $(x_n, f(x_n))$. It fails whenever $f(x_{n-1}) = f(x_n)$ (Figure 2b).

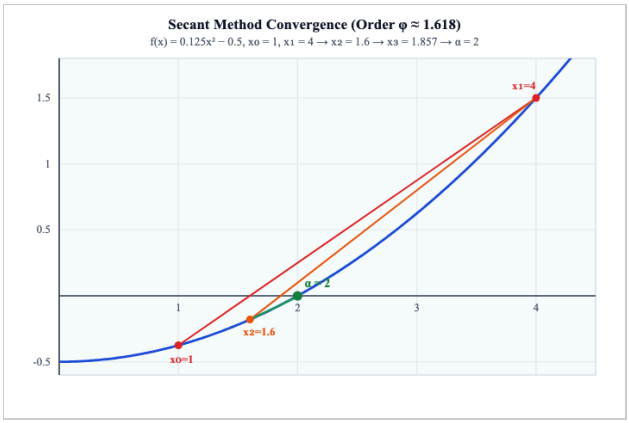


Figure 2a. Secant method convergence from $(x_0, x_1) = (1, 4)$.

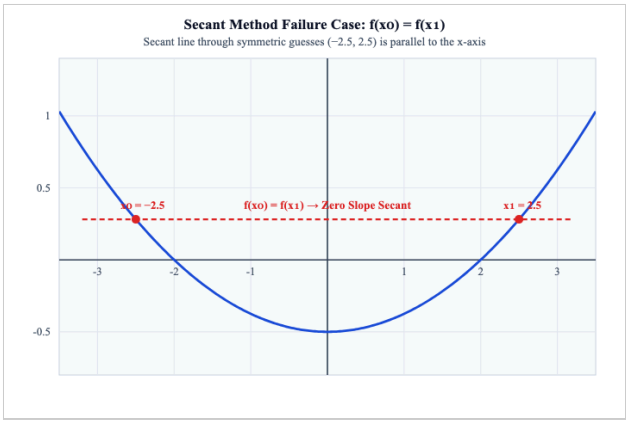


Figure 2b. Secant method failure when $f(x_0) = f(x_1)$.

Table 2. Secant Method Iterations for $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$ ($x_0 = 1, x_1 = 4$).

Step n	x_n	x_{n+1}	α	$e_n = x_n - \alpha$	$f(x_n)$
0	1.000000	4.000000	2	-1.000000	-0.375000
1	4.000000	1.600000	2	2.000000	1.500000
2	1.600000	1.857143	2	-0.400000	-0.180000
3	1.857143	2.016529	2	-0.142857	-0.068878
4	2.016529	1.999390	2	0.016529	0.008299
5	1.999390	1.999997	2	-0.000610	-0.000305
6	1.999997	2.000000	2	-0.000003	-0.000001

1.3. Bisection Method

The Bisection method maintains a sign-changing bracket $[l_n, r_n]$ with $f(l_n)f(r_n) < 0$ and halves the interval at each step via $x_n = (l_n + r_n)/2$.

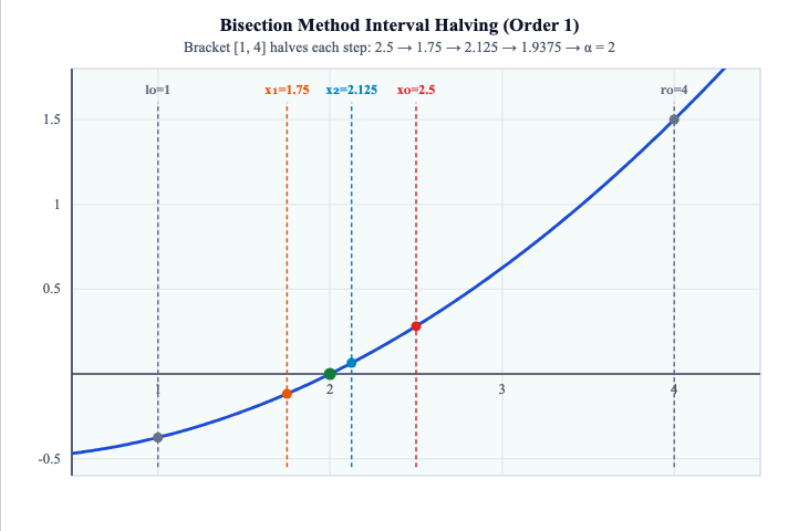


Figure 3. Bisection interval halving on $[1, 4]$ for $f(x) = \frac{1}{8}x^2 - \frac{1}{2}$.

Definition 1.1 (Order of Convergence). *A sequence x_n converging to α has order of convergence $q \geq 1$ if $\lim_{n\to\infty} \frac{|x_{n+1}-\alpha|}{|x_n-\alpha|^q} = \mu$ for some asymptotic constant $\mu > 0$.*

2. Polynomials, Root Bounds, and Critical Points

Under an affine transformation $q(x) = c \cdot p(ax + b)$ ($a, c \neq 0$), every root α of p corresponds to a root $(\alpha - b)/a$ of q. Normalizing by the leading coefficient a_n yields Cauchy's bound on all root and critical-point magnitudes.

Proposition 2.1 (Cauchy Bound on Root Magnitudes). *Given a polynomial* $p(x) = \sum_{r=0}^{n} a_r x^r$ *with* $a_n \neq 0$*, all roots of* p *lie strictly within* $[-M_p, M_p]$*, where*

$$M_p = \max\left\{1, \sum_{r=0}^{n-1} \left|\frac{a_r}{a_n}\right|\right\}.$$

Proof. Setting $b_r = a_r/a_n$, any root α satisfying $|\alpha| \geq 1$ obeys $|\alpha|^n = |\sum_{r=0}^{n-1} b_r \alpha^r| \leq |\alpha|^{n-1} \sum_{r=0}^{n-1} |b_r|$ by the triangle inequality. Dividing by $|\alpha|^{n-1}$ yields $|\alpha| \leq \sum_{r=0}^{n-1} |a_r/a_n| \leq M_p$. ∎

Corollary 2.2. *All critical points of* $p(x)$ *(roots of* $p'(x)$*) are also bounded within* $[-M_p, M_p]$*, since* $M_{p'} = \max\left\{1, \sum_{r=1}^{n-1} \left|\frac{r a_r}{n a_n}\right|\right\} \leq M_p$.

Proposition 2.3 & Corollary 2.4 (Critical-Point Separation). *By Rolle's Theorem, there is always a critical point between any two distinct real roots of* $p(x)$*. Consequently, there can be at most one real root strictly between any two consecutive critical points (CCPs)* (c_1, c_2)*. Moreover, if* $\alpha \in (c_1, c_2)$ *is a simple root,* $p(x)$ *and* $p'(x)$ *have opposite signs on* $[c_1, \alpha)$ *and identical signs on* $(\alpha, c_2]$.

Lemma 2.5 (Multiplicity Characterization). *A value* α *is a root of* $p(x)$ *with multiplicity* $k \geq 1$ *(*$p^{(l)}(\alpha) = 0$ *for* $0 \leq l < k$ *and* $p^{(k)}(\alpha) \neq 0$*) if and only if* $p(x) = (x-\alpha)^k q(x)$ *with* $q(\alpha) \neq 0$.

3. Newton's Method & Inflection-Guided Basins

Definition 3.1. *Newton's rational function is* $\Psi_p(x) = x - \frac{p(x)}{p'(x)}$*, with* n*-fold iterate* $\Psi_p^n(x) = \Psi_p(\Psi_p^{n-1}(x))$.

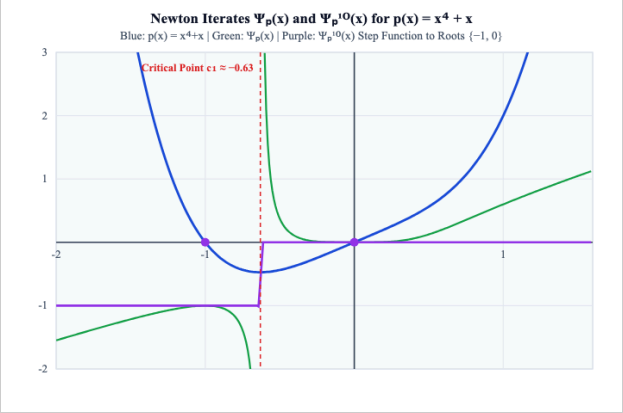


Figure 4. $p(x) = x^4 + x$ (blue), $\Psi_p(x)$ (green), and $\Psi_p^{10}(x)$ (purple).

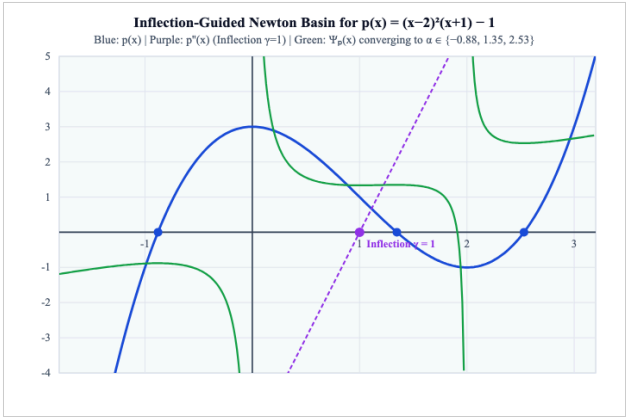


Figure 5. $p(x) = (x-2)^2(x+1) - 1$, $p''(x)$ (purple), and $\Psi_p(x)$ (green).

Lemma 3.2 & Proposition 3.3 (Monotone Basin Existence). *If p is monotonic and Ψ_p is monotonically increasing on $[a,b]$ containing a simple root α, then $\forall x_0 \in (a,b)$, $\Psi_p^n(x_0) \to \alpha$ monotonically. Furthermore, partitioning any sign-changing CCP bracket $[c_\alpha, c'_\alpha]$ by inflection points $p''(\gamma_i) = 0$ isolates a sub-interval $[\gamma_t, \gamma_{t+1}]$ containing α where $\operatorname{sign}(p(x_0)) = \operatorname{sign}(p''(x_0))$ at one endpoint x_0, guaranteeing $\Psi'_p(x) = \frac{p(x)p''(x)}{(p'(x))^2} > 0$.*

Algorithm 1: Inflection-Guided Initial Guess for Newton's Method

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1. **Input:** Polynomial $p(x)$, consecutive critical points $c_1 < c_2$ with $p(c_1)p(c_2) < 0$, sorted inflection points $D = (\gamma_1, \ldots, \gamma_r) \subset (c_1, c_2)$.
2. Let $s = \operatorname{sign}(p'((c_1+c_2)/2))$ and form partition $P = [c_1, \gamma_1, \ldots, \gamma_r, c_2]$.
3. For each adjacent pair (a,b) in P with $p(a)p(b) \le 0$: if $\operatorname{sign}(p''((a+b)/2)) = s$, return $x_0 = b$; else return $x_0 = a$.

Theorem 3.4 (Order of Convergence of Newton's Method). *When $x_n = \Psi_p^n(x_0) \to \alpha$ for a root α of multiplicity $k \ge 1$, the error $e_n = x_n - \alpha$ satisfies*

$$e_{n+1} = e_n \left(1 - \frac{1}{k}\right) + e_n^2 \frac{p^{(k+1)}(\alpha)}{k(k+1)p^{(k)}(\alpha)} + \mathcal{O}(e_n^3).$$

Thus Newton's method converges quadratically ($q = 2$) with $\mu = \left|\frac{p''(\alpha)}{2p'(\alpha)}\right|$ when $k = 1$, and linearly ($q = 1$) with rate $1 - 1/k$ when $k > 1$.

4. Secant Method & Bisection Method

Theorem 4.1 (Golden-Ratio Convergence of the Secant Method). *If the secant sequence $x_{n+1} = x_n - f(x_n)\frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}$ converges to a simple root α with $f'(\alpha), f''(\alpha) \neq 0$, its order of convergence is the golden ratio $\phi = \frac{1+\sqrt{5}}{2} \approx 1.618034$.*

Proof. Expanding $f(x_n)$ and $f(x_{n-1})$ via Taylor series around α gives $e_{n+1} \approx C e_n e_{n-1}$ where $C = \frac{f''(\alpha)}{2f'(\alpha)}$. Setting $g_n = \log|Ce_n|$ yields the Fibonacci recurrence $g_{n+1} \approx g_n + g_{n-1}$, so $\lim_{n\to\infty} \frac{g_{n+1}}{g_n} = \phi = \frac{1+\sqrt{5}}{2}$ and $\lim_{n\to\infty} \frac{|e_{n+1}|}{|e_n|^\phi} = |C|^{\phi-1}$. ∎

Theorem 4.2 (Linear Convergence of Bisection). *On any bracket $[l_0, r_0]$ with $f(l_0)f(r_0) < 0$, bisection satisfies $r_n - l_n = (r_0 - l_0)/2^n$ and converges with order $q = 1$ and rate $\mu = 1/2$.*

5. Polynomial Factorization Procedure (PFP)

Combining Proposition 2.1 (Cauchy bound M_p), Lemma 2.5 (multiple roots occur at critical points), and Algorithm 1 (inflection-guided Newton initialization), we obtain the complete recursive **Polynomial Factorization Procedure (PFP)**.

Algorithm 2: Polynomial Factorization Procedure (PFP)

1. **Base Case:** If $\deg(f) \in \{1, 2\}$, solve for real roots and multiplicities directly via closed-form linear/quadratic formulas.
2. **Recursive Derivative Calls:** Compute critical points $\{(c_i, m_i)\} = \mathrm{PFP}(f')$ and inflection points $\{d_j\} = \mathrm{PFP}(f'')$.
3. **Multiple Root Detection:** For each (c_i, m_i), if $|f(c_i)| = 0$, record c_i as a root of f with multiplicity $m_i + 1$.
4. **Simple Root Isolation:** Partition $[-M_f, M_f]$ into monotone intervals $(-M_f, c_1), (c_1, c_2), \ldots, (c_k, M_f)$. On each sign-changing interval, select x_0 via Algorithm 1 using inflection points D_j and run safeguarded Newton-Bisection to converge to the unique simple root α (multiplicity 1).

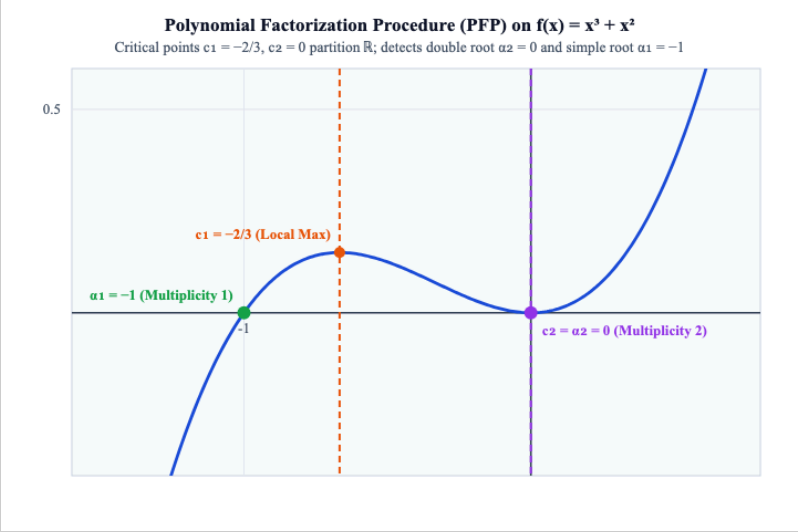


Figure 6. PFP partitioning $f(x) = x^3 + x^2$ by critical points $c_1 = -2/3$ and $c_2 = 0$, detecting double root $\alpha_2 = 0$ ($m = 2$) and simple root $\alpha_1 = -1$ ($m = 1$).

Theorem 5.1 (Completeness of PFP). *For any real polynomial $f(x)$, the Polynomial Factorization Procedure isolates and approximates all real roots along with their exact multiplicities.*

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